

Figures, Tables, and Presentation Prep

ECON 692 – Applied Economics Seminar

Andrew Hobbs

University of San Francisco

April 16, 2026

Today's Agenda

Time	Duration	Activity
4:35	10 min	Check-in: where are you on the draft?
4:45	25 min	What makes a good figure (and table)
5:10	10 min	Break
5:20	20 min	Group brainstorm: key figures for each project
5:40	60 min	Solo work sprint: build your figures
6:40	25 min	Reconvene: share figures, get feedback
7:05	15 min	Presentation prep: slide design
7:20	25 min	Work sprint 2
7:45	10 min	Wrap-up & next steps

Check-in

Full Draft is due April 24 — one week from today.

Quick poll:

- Who has at least one polished figure?
- Who has a regression table?
- Who has started building presentation slides?

Today's goal: leave with your key figures drafted and a plan for your presentation.

What Makes a Good Figure

What Figures Does Your Project Need?

The answer depends on your method:

DiD / Event Study

- Event study plot (the money figure)
- Pre-trends test
- Raw outcome trends by group

Synthetic Control / SDID

- Treated vs. synthetic outcome
- Gap plot (treatment effect over time)
- Placebo tests

IV

- First-stage scatter / binned scatter
- Reduced-form relationship

Everyone

- Descriptive time series or map
- Summary statistics table
- Distribution of key variables

Anatomy of a Good Figure

Every figure needs these elements:

1. **Informative title.** Not “Figure 1” — tell the reader the takeaway.
2. **Labeled axes with units.** “Log earnings (\$2020)” not just “Y.”
3. **Direct labels on lines.** Almost always better than a legend box.
4. **Source note.** Where the data come from, sample size, filters applied.
5. **Confidence intervals.** If you're plotting estimates, show uncertainty.

A figure should be interpretable without reading the surrounding text.

Design Principles: Declutter and Annotate

Declutter

- Remove gridlines (or make them very light)
- Drop the border/box around the plot
- No 3D effects, ever
- Maximize the “data-ink ratio”

Color with Purpose

- Use color to distinguish, not decorate
- Check colorblind accessibility (viridis, okabe-ito)
- Gray out everything that isn't the main message

Annotate

- Mark key events directly on the plot
- Label lines directly instead of using a legend
- Add reference lines (zero effect, mean, threshold)

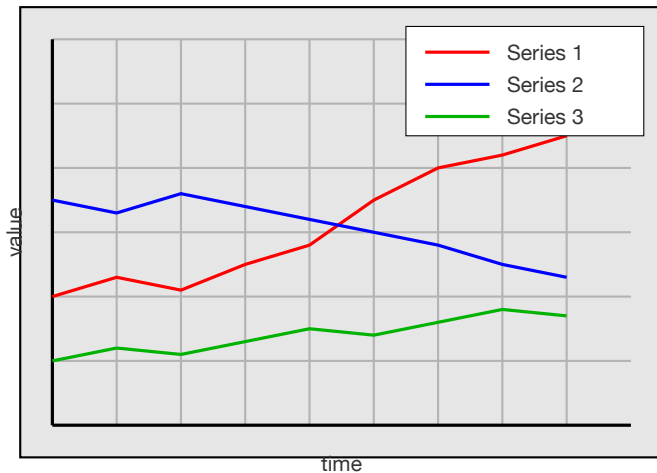
One Message Per Figure

- If you're showing two things, make two figures
- The reader should get the point in 5 seconds

Common Figure Mistakes

- **Chartjunk.** Unnecessary gridlines, borders, backgrounds, 3D effects. Strip it all out.
- **Unlabeled or cryptic axes.** “var_emp_q3_adj” is not a label. Write “Employment (thousands).”
- **Too many series.** Eight lines on one plot? Highlight 1–2 and gray out the rest.
- **Misleading scales.** Truncated y-axes, dual axes, inconsistent scales across panels.
- **Default everything.** R and Python defaults are a starting point, not a final product.
- **No annotation of key events.** Policy change at time t ? Draw a vertical line and label it.

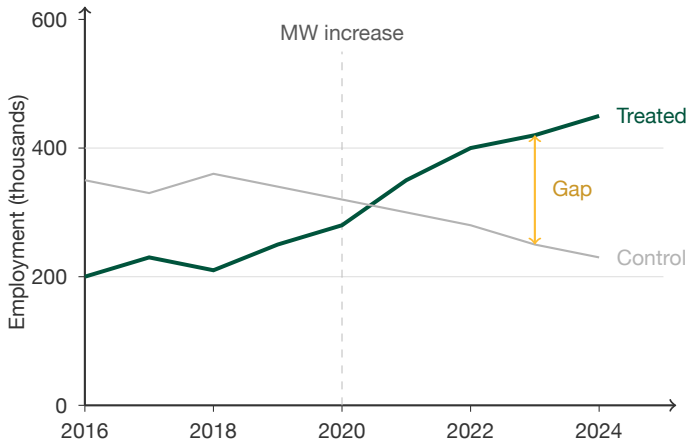
Before: The Cluttered Chart



Gray background, heavy gridlines, rainbow colors, useless labels, legend box covering data.

After: The Clean Chart

Teen employment fell sharply after the 2020 minimum wage increase



Source: CPS monthly data, ages 16–19. $N = 48,000$.

What Changed?

Before

- Gray background, thick gridlines
- Legend box covering data
- Axis labels: “value” and “time”
- Rainbow color palette
- No title, no source note
- Three unnamed “series”

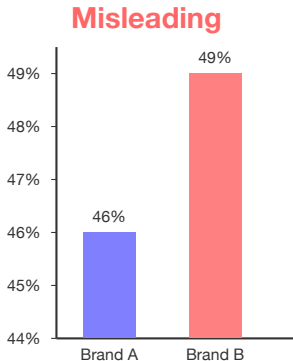
After

- White background, minimal gridlines
- Direct labels on lines
- “Employment (thousands)”
- Two purposeful colors + annotation
- Title states the finding
- Source note with sample info

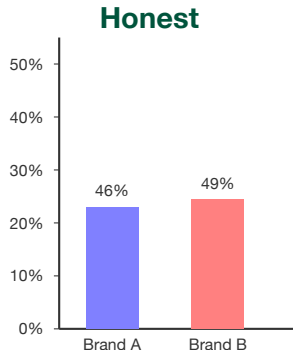
The data are identical. The difference is **design**. Use `theme_minimal()` as a starting point, then customize from there.

Misleading: The Truncated Y-Axis

A 3-percentage-point difference made to look enormous.



Brand B looks $2.5\times$ taller!

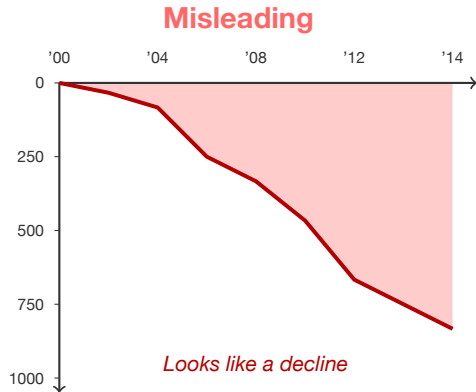


The real difference: 3 points.

Tufte called this the **“lie factor”**: the ratio of the visual effect to the data effect. If the axis doesn’t start at zero for a bar chart, ask why.

Misleading: The Inverted Axis

Inspired by a real Reuters graphic on Florida gun deaths.

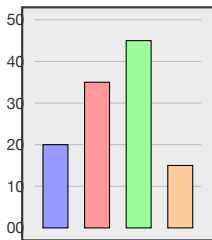


Same data. Flipping the y-axis made a **dramatic increase** look like a decline. Always check axis orientation.

The Data-Ink Ratio

Tufte's principle: maximize the share of ink that shows **data**. Erase everything else.

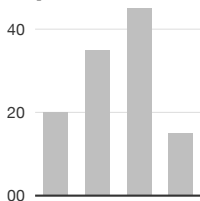
Step 1: Default



Background, borders,

gridlines, 4 colors

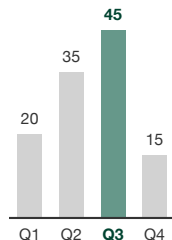
Step 2: Remove



White background,

fewer gridlines, one color

Step 3: Label



Direct labels, highlight

the message, no gridlines

Data-ink ratio = data ink / total ink. Each step removes non-data ink without losing information.

Tables

Regression Table Essentials

A good regression table has:

1. **Clear column headers.** Each column = one specification. Label what changes: “+ Controls,” “+ FE,” “Alt. Sample.”
2. **Readable variable names.** Not `log_inc_cpi_adj`. Write “Log income (2020 \$).”
3. **Standard errors.** In parentheses, with a note on clustering level.
4. **Stars — but don't overdo it.** One system ($*p < 0.05$), applied consistently.
5. **N, R^2 , FE indicators.** At the bottom. Use Yes/No rows for fixed effects.
6. **Only show key coefficients.** Don't report every control. The reader cares about your treatment effect.
7. **Notes.** Clustering, sample restrictions, variable definitions.

Tools: `etable()` in `fixest`, `modelsummary()`, `stargazer()`.

Group Brainstorm

Activity: Figure Brainstorm

Form groups of 3–4.

Round 1 (2 min each): Briefly describe your project — question, method, data.

Round 2 (5 min each): As a group, brainstorm the key figures for each person's project. For each figure, write down:

- What it shows (one sentence)
- What goes on each axis
- Why it matters for the argument

Aim for **3–5 figures per project**. You don't need all of them — but you need to know which ones matter most.

Guiding Questions

As you brainstorm, ask:

1. **What's the one figure a reader remembers?** The event study, gap plot, or scatter that *is* your result.
2. **What shows your identification works?** Pre-trends, balance tests, first-stage strength, placebos.
3. **What descriptive fact motivates the question?** A trend, a map, a distribution.
4. **What does the skeptic want to see?** Robustness, heterogeneity, alternative specifications.

Rank your list. Start building from the top.

Work Sprint

Work Sprint 1 (5:40–6:40)

Build your figures. Start with the most important one.

Priority order:

1. **Main result figure.** Event study, SC plot, IV scatter — whatever shows your key finding.
2. **Identification figure.** Pre-trends, balance, first stage — whatever builds credibility.
3. **Descriptive motivation.** The fact or pattern that makes the reader care.

Tips: Start with the simplest version — get data on the plot first, then polish. Use `theme_minimal()` as your base. Save as PDF or PNG at high resolution.

Reconvene

Reconvene: Share Your Figures (6:40–7:05)

Back in your groups. Each person: show 1–2 figures.

Feedback protocol:

1. **First impression.** Look at the figure for 10 seconds without explanation. What do you think it shows?
2. **Clarity check.** Can you read the axes? Do you know the units? Is the takeaway obvious?
3. **What's missing?** Labels, annotations, confidence intervals, reference lines, title?
4. **Does it support the argument?** Is this the right figure for the point being made?

If your group can't tell what the figure shows without your explanation, that's the most useful feedback you'll get.

Presentation Prep

Final Presentations: The Setup

- **May 7:** First group (~3 presenters, 1 hour)
- **May 14:** Remaining presenters (full class period)
- **About 15 minutes each** including Q&A

Target: ~**12–15 slides** for 12–13 minutes of talking, plus 2–3 minutes for questions.

We'll finalize the schedule and order next week.

Pick Your Date

Please fill out the survey so we can set the schedule:

- **May 7:** ~3 slots (first hour of class)
- **May 14:** ~10 slots (full period)
- Either date works? Say so — it helps with scheduling.

tinyurl.com — or just scan the QR code →



Presentation Date Survey

Slide Design Principles

1. **One idea per slide.** If you need two bullet lists, make two slides.
2. **Assertion–evidence structure.** The slide title is your claim: “Minimum wage increases reduced teen employment by 3%.” The body is the evidence: the figure or table that proves it.
3. **More figures, less text.** Your slides aren’t a script. Use them to show evidence; you provide the narration.
4. **Big fonts, big figures.** If someone in the back row can’t read it, make it bigger.
5. **Build the narrative.** Motivation → question → data → method → results → so what. Same arc as the project, just compressed.

Presentation Structure Template

A 15-minute economics presentation, slide by slide:

1. Title slide
2. Motivation (the hook)
3. Research question
4. Related work (1 slide max)
5. Data overview
6. Summary statistics
7. Empirical strategy
8. Identification (why it's credible)
9. Main result (the money slide)
10. Robustness / mechanisms
11. Interpretation & implications
12. Conclusion (1 slide)

That's 12 slides. Spend most of your time on 7–10.

Common mistake: spending 8 minutes on background and 4 on results. Flip that.

Work Sprint 2

Work Sprint 2 (7:20–7:45)

Pick your priority:

Figures Track

1. Apply feedback from your group
2. Polish labels, colors, annotations
3. Start your second or third figure

Slides Track

1. Set up your slide deck (Google Slides, Beamer, Quarto)
2. Build your title + motivation slides
3. Drop in existing figures

Whatever you don't finish today, do before next week.

Wrap-up

Next Week & Deadlines

Next week (4/23): Peer review workshop and presentation practice.

Upcoming:

- Weekly progress report due **Wednesday**
- **Full Draft due April 24** (one week)
- Final presentations: May 7 & May 14

Between Now and the Full Draft

Your full draft needs: intro, data description, empirical strategy, results with figures and tables, and conclusion. Leave today with your key figures built, and you're in good shape.